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A highly hydrophilic cation exchange nonwoven with a further modifiable epoxy group prepared by radiation-induced graft polymerization

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Novel cation exchange nonwoven PP-*g*-SSS/GMA containing epoxy and sulfonic groups was successfully prepared *via* radiation-induced simultaneous grafting polymerization by attaching glycidyl methacrylate (GMA) and styrene sulfonic sodium (SSS) monomers onto nonwoven polypropylene (PP). The ion exchange capacity of the obtained PP-*g*-SSS/GMA reached more than 1.5 mmol g⁻¹ and the content of the epoxy groups with potential modifiable properties was above 3 mmol g⁻¹. The relative content of the two functional groups could be adjusted *via* changing the initial monomer concentration ratio. Scanning electron microscope (SEM) images showed that the polymer surface was fully covered with grafts after modification and those amorphous grafts diluted the crystallinity of the matrix, as revealed *via* differential scanning calorimetry (DSC). Attenuated total reflection spectroscopy (ATR) confirmed the successful grafting of GMA and SSS onto PP nonwovens. Water contact angle measurements showed that the contact angle decreased from 100.94° for the PP matrix to 0° for PP-*g*-SSS/GMA with a 56% degree of grafting, indicating the outstanding hydrophilicity of PP-*g*-SSS/GMA upon the introduction of SSS. Besides this, the epoxy groups on PP-*g*-SSS/GMA could be further modified *via* reacting them with chemical compounds with selective adsorption groups. Ag(I) adsorption tests proved that PP-*g*-SSS/GMA@EDA obtained *via* the ethanediamine (EDA) modification of epoxy groups on PP-*g*-SSS/GMA shows excellent selectivity towards Ag(I), and the hydrophilicity brought about by sulfonic groups also ensured a fast adsorption rate. The novel cation exchange nonwoven PP-*g*-SSS/GMA fabric has potential applications in water treatment, as its strong hydrophilicity could speed up the diffusion rates of adsorbed ions, and further functionalization of the epoxy groups could enable it to selectively adsorb ions in aqueous solutions.

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1. Introduction

Since the last century, ion exchange materials as a new type of material have been widely studied because of their applications in a variety of fields, including, but not limited to, electrodialysis, diffusion dialysis, and the recovery of acid or alkali from waste acid or alkali solutions. Ion exchange materials are generally composed of hydrophobic substrates, immobilized ion-functionalized groups, and movable counter-ions.¹ Nonwoven polypropylene (PP) is one of the potential support materials for use in separation applications because of its excellent mechanical properties, chemical inertness, processability, and low cost. However, commercial PP is a hydrophobic polymer, which severely limits its wide application in aqueous solutions.^{2,3} Another important limitation of nonwoven PP is

its chemical inertia due to it having no reactive side-chain groups, making surface and bulk modifications a challenging task. The research on the modification of nonwoven PP has been devoted to improving its hydrophilicity or the addition of new functionalities.^{4,5}

Generally, chemically initiated treatments use organic solvents and generate waste, which could be potentially harmful to the environment. And the inert surface of nonwoven PP increases the difficulty of its chemical modification.^{3,6,7} Instead, irradiation-induced methods based on gamma(γ)-rays, plasma, and electron beam are preferred in many cases, because apart from being environmentally friendly, a permanent and decent grafted layer can form on the nonwoven surface without degrading its original characteristics.^{7–9} The quantitative *in situ* free radicals (active sites) generated by irradiation can initiate the grafting of functional monomers on the PP surface, imparting the raw materials with both physically and chemically desirable properties.^{10,11} γ -Ray-induced

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grafting functionalization is a useful technique for the controlled surface modification of a broad range of polymers.

The sulfonic acid group ($-\text{SO}_3\text{H}$) is a typical functional group for cation exchange material modification.¹² Among the common monomers containing $-\text{SO}_3\text{H}$, styrene sulfonic sodium (SSS) is more suitable for radiation graft because of its unsaturated bonds ($\text{C}=\text{C}$). SSS has an outstanding function in the modification of materials, as it can not only reverse the hydrophobicity of modified matrixes, but the introduction of movable charged ions (Na^+) also makes the matrixes ion-exchangeable.¹³ However, ion exchange materials exhibit low selectivity among cations or anions.^{14,15} Ion-exchangers containing a single functional group (such as sulfonic acid type adsorbents) cannot satisfy the selective adsorption of specific ions or substances. The main adsorption mechanism of a sulfonic acid adsorbent is ion exchange, which is affected by the charge and radius of the adsorbed ions. And there is another adsorption mechanism supporting the selectivity of the adsorbent, which is hard-soft acid base theory, *e.g.*, mercapto (sulfur as a soft donor) and amine (nitrogen as a borderline donor atom) groups possess affinity for $\text{Ag}(\text{I})$.¹⁶

Glycidyl methacrylate (GMA) has two groups (epoxy and carbonyl groups), and can be used as a medium to further covalently or noncovalently link with molecules required for particular applications, such as heavy metal recovery.^{17,18} Toledo *et al.* synthesized a polymer, *N*-methyl-D-glucamine (P-(NMG)), using glycidyl methacrylate-*N*-methyl-D-glucamine (GMA-NMG) as the monomer *via* radical polymerization and after combination with ultrafiltration membranes, where P-(NMG) showed potential as an alternative for the removal of hazardous $\text{As}(\text{V})$ species from aqueous solutions at acidic pH.¹⁹ In our past work, we used GMA as the intermediate, immobilized mercaptoethylamine (MEA) onto nonwoven PP and synthetic PP-GMA@MEA was successfully applied to the selective adsorption of $\text{Ag}(\text{I})$ under highly acidic conditions.²⁰ In brief, GMA can further improve the applicability of materials.

In summary, enhanced practicability in water treatment with nonwoven PP will likely require an increase in hydrophilicity and selectivity, so the synthesis of novel nonwoven PP-based materials that exhibit hydrophilicity, cation exchange and selective adsorption properties is a topic worth studying. Up to now, the synthesis of cation exchange materials based on a PP matrix *via* simultaneous irradiation-grafting with the functional monomers GMA and SSS has not been reported. In this work, we report a new method to synthesize the novel cation exchange nonwoven PP-*g*-SSS/GMA material in one step. In this synthesis, two functional monomers, SSS and GMA, were introduced onto PP nonwovens *via* γ -ray-induced simultaneous radiation graft polymerization, and the effect of the monomer content on grafting was investigated. The physical, chemical, thermodynamics, and other properties of the nonwovens were investigated using various characterization methods. By modifying PP-*g*-SSS/GMA with EDA, PP-*g*-SSS/GMA@EDA was obtained. As an adsorbent, the $\text{Ag}(\text{I})$ adsorption performance of the modified material from water was

investigated to verify the potential application value of PP-*g*-SSS/GMA in water treatment.

2. Experimental

2.1 Materials and chemicals

Hydrophobic PP nonwovens with a density of 0.094 g cm^{-3} and 1 mm in thickness used in this study were purchased from Changtai Co. (Shenzhen, Guangdong, China). The commercial resin IRN77 was obtained from Acros Organics (Belgium). Glycidyl methacrylate (GMA, 97%) and styrene sulfonic sodium (SSS, 90.0%) was purchased from Macklin (Shanghai, China). *N,N*-dimethyl acrylamide (DMF, 99.5%), methanol (99.5%), tetrahydrofuran (99%), ethanediamine (EDA, 99%), cobalt nitrate hexahydrate (99%) and strontium nitrate (99%) were purchased from Aladdin (Los Angeles, CA, USA). Copper sulfate pentahydrate (98%) was purchased from Bidepharm (Shanghai, China). Nitric acid, silver nitrate (99%) and nickel nitrate hexahydrate (98%) were obtained from Sinopharm Chemical Reagent Co. (Shanghai, China). Manganous nitrate (50 wt% in H_2O) was purchased from Energy Chemical (Shanghai, China). All the reagents used were analytical grade. The water used throughout the study was deionized.

2.2 Manufacturing of PP-*g*-SSS/GMA

To begin with, DMF and water at a volumetric ratio of 1:2 were mixed together as a solvent at room temperature. Then, GMA and SSS as monomers were added into glass reagent bottles with the prepared solvent under continuous stirring to make a homogeneous solution. The content of SSS was fixed at 0.50 mol L^{-1} , while the volume fraction (v/v) of GMA was varied from 0% to 13%, corresponding to the molar concentration range of 0 to 0.957 mol L^{-1} . Also, a polymerization inhibitor (copper sulfate pentahydrate) was added to the solution at a volume proportion of 0.5%; copper sulfate solution with a higher volume concentration (5%) was prepared first, and then this was added at a ratio of 1:100 by volume to obtain a concentration of 0.5‰. After purging with nitrogen gas for 30 min, the PP nonwovens were immersed in glass reagent bottles and soaked overnight, followed by another 30 min of purification with nitrogen. The prepared samples were irradiated with γ -rays (Shanghai Heming Radiation Science & Technology Co., Ltd, China; $1.11 \times 10^{16} \text{ Bq}$; 1.04 kGy h^{-1}). After irradiation, the grafted PP nonwovens were washed with DMF and then repeated heat soaking and pumping filtration in a mixture of methanol and tetrahydrofuran were used to remove the homopolymer.²¹

To measure their degree of grafting (DG), each nonwoven was prepared at 60°C for 24 h in a vacuum oven. The DG was then calculated using the following equation:⁹

$$\text{DG}(\%) = \frac{w_1 - w_0}{w_0} \times 100 \quad (1)$$

where w_0 and w_1 are the dry weights of the PP nonwovens before and after grafting, respectively.

2.3 Characterization

Morphological and component analyses on the surface of pristine PP and PP-*g*-SSS/GMA were carried out by scanning electron microscopy (SEM) and energy-dispersive X-ray spectroscopy (EDS) using a NOVA Nano SEM 230 microscope (FEI, USA) equipped with energy-dispersive X-ray apparatus.

To investigate the differences in the chemical structures of pristine and modified PP nonwovens, the attenuated total reflection (ATR) spectra of pristine PP and PP-*g*-SSS/GMA nonwovens were obtained on a Nicolet iS5 transform infrared reflection spectrometer (Thermo Nicolet Company, USA) in the range of 4000 to 600 cm⁻¹ at a resolution of 4 cm⁻¹ over 32 scans.

The ion exchange capacity (IEC) of each test set was measured using a titration method.^{13,22} Each nonwoven was immersed in 1.0 mol L⁻¹ HCl solution for 24 h to convert Na⁺ into H⁺. Following this, after removal of the grafted PP nonwoven samples, the solution was titrated with 0.01 mol L⁻¹ NaOH using an 888 Titrand instrument (Metrohm, Swit). The IEC (mmol g⁻¹) and epoxy group content (EC, mmol g⁻¹) of the nonwovens were calculated using eqn (2) and (3),⁹ respectively. Furthermore, according to the known IEC and EC, the DG of SSS and GMA can be further deduced using eqn (4) and (5), so as to reveal the grafting results of each monomer.

$$\text{ICE} = \frac{C_m V_m - C_n V_n}{w_1} \quad (2)$$

$$\text{EC} = \frac{w_1 - w_0 - \text{IEC} \times 206.19 \times w_1}{142.15 \times w_1} \quad (3)$$

$$D_{\text{SSS}}(\%) = \frac{\text{IEC} \times w_1 \times 206.19}{w_0 \times 1000} \times 100 \quad (4)$$

$$D_{\text{GMA}}(\%) = \frac{\text{EC} \times w_1 \times 142.15}{w_0 \times 1000} \times 100 \quad (5)$$

where w_0 and w_1 are the dry weights of the PP nonwovens before and after grafting, V and C are the volume and concentration of HCl(m) and NaOH(n), and 206.19 and 142.15 are the molar masses of SSS and GMA.

Water contact angles of nonwovens were analyzed using the sessile-drop technique, performed on a JY-82 standard goniometer (Dingsheng, China) equipped with a high-speed digital camera that operates at 100 fps.

Thermogravimetric analysis (TGA) was used to study the thermal stability of pristine PP and grafted PP-*g*-SSS/GMA nonwovens. TGA was performed on a Pyris1 thermogravimetric analyzer (PerkinElmer, USA) under a nitrogen gas flow of 20 mL min⁻¹ at a heating rate of 10 °C min⁻¹. And the TGA data was recorded on DTG-60A (Shimadzu, Japan) apparatus under and air atmosphere of 20 mL min⁻¹ at a heating rate of 10 °C min⁻¹.

Differential scanning calorimetry (DSC) was performed on DSC-2500 (TA Instruments, USA) apparatus under a nitrogen gas flow of 20 mL min⁻¹ at a heating rate of 10 °C min⁻¹. The degree of crystallinity (DC) of each irradiated sample was calculated using eqn (6). The melting enthalpy of PP com-

ponents in the grafted samples ($\Delta H_{\text{f(PP)}}$) was calculated using eqn (7).

$$\text{DC}(\%) = \Delta H_{\text{f(1)}} / \Delta H_{\text{f(0)}} \times 100 \quad (6)$$

$$\Delta H_{\text{f(PP)}} = \frac{\Delta H_{\text{f(1)}}}{1/(1 + \text{DG})} \quad (7)$$

where $\Delta H_{\text{f(0)}}$ is the melting enthalpy of PP with 100% crystallinity, 207 J g⁻¹,²³ and $\Delta H_{\text{f(1)}}$ is the melting enthalpy of the samples.

To measure the mechanical properties of the materials, tensile test samples were tested on a CMT6103 tensile testing machine (MTS, USA) with a cross-head speed of 1 mm min⁻¹ at room temperature.

2.4 Further modification and adsorption tests towards Ag(I)

PP-*g*-SSS/GMA-86 with 1.28 mmol g⁻¹ of EC and 1.36 mmol g⁻¹ of IEC was further modified by the reaction between the epoxy groups in GMA and EDA, and the composite was named PP-*g*-SSS/GMA@EDA. Ag(I) adsorption studies, in terms of both adsorption selectivity and adsorption kinetics, of the synthesized product PP-*g*-SSS/GMA@EDA were carried out experimentally. In order to intuitively illustrate the advantages of the bifunctional copolymer PP-*g*-SSS/GMA, the same EDA-modification reaction was carried out on PP-*g*-GMA using 1.28 mmol g⁻¹ synthesized in the previous study²⁰ to obtain PP-*g*-GMA@EDA, and then comparative adsorption studies were conducted. Another contrast adsorbent in the adsorption test was the commercial cationic resin IRN77 with a single functional group, a sulfonic group.

2.4.1 EDA-modification of PP-*g*-SSS/GMA and PP-*g*-GMA. An EDA-modification reaction was performed by immersing PP-*g*-SSS/GMA or PP-*g*-GMA in 70% (w/w) EDA/isopropanol solution at 70 °C (ref. 24) over 6 h. The obtained products were washed with water and methanol to removal any residual solvent and EDA, followed by overnight drying in a vacuum oven at 40 °C. The EDA content (mmol g⁻¹) on PP-*g*-SSS/GMA@EDA and PP-*g*-GMA@EDA was calculated according to eqn (8) and the conversion percentage (%) of epoxy groups with EDA was determined using eqn (9).

$$\text{EDA content} = \frac{w_2 - w_1}{60.1 \times w_1} \times 1000 \quad (8)$$

$$\text{conversion percentage} = \frac{\text{EDA content}}{\text{EC}} \times 100 \quad (9)$$

where w_1 and w_2 are the dry weights of PP-*g*-SSS/GMA or PP-*g*-GMA and PP-*g*-SSS/GMA@EDA or PP-*g*-GMA@EDA, 60.1 is the molar mass of EDA, and EC can be obtained according to eqn (3).

2.4.2 Adsorption tests towards Ag(I). The adsorption of Ag(I) from aqueous solution was performed in batch experiments using PP-*g*-SSS/GMA@EDA, PP-*g*-GMA@EDA, and IRN77 as adsorbents at room temperature (25 °C). The effect of solution acidity (pH) adjusted by HNO₃ on the adsorption capacity was investigated, where 10 mg of adsorbent was placed in 10 ml of a 2 mmol L⁻¹ AgNO₃ solution shaken at 100 rpm for 24 h. The

effect of contact time on the adsorption process was also studied. In this procedure, 10 mg of adsorbent was added to 10 ml of a 2 mmol L⁻¹ AgNO₃ solution and 50 μL of solution was taken out at different times to analyze the Ag concentration as shaking was maintained. The selectivity of the adsorbents was also investigated, where mixed solutions of 1 mmol L⁻¹ Ag(I), Mn(II), Co(II), Ni(II) and Sr(I) at different pH values were prepared first, and then 10 mg of adsorbent was added to the metal mixture solution and the shaking of the table was maintained at 100 rpm for 24 h. The concentrations of the metal ions before and after adsorption process were analyzed using an ICPS-7510 instrument (Shimadzu, Japan). All of the batch experiments were repeated three times. According to eqn (10), the adsorption capacity (Q_e , mmol g⁻¹) of adsorbents towards metal ions was calculated. The separation factor (SF) was determined using eqn (11).

$$Q_e = \frac{(C_0 - C_e) \times V}{m} \quad (10)$$

$$SF = \frac{k_i}{k_j} = \frac{(Q_e/C_e)_i}{(Q_e/C_e)_j} \quad (11)$$

where C_0 and C_e (mmol L⁻¹) are the ion concentrations before and after adsorption; V (ml) is the volume of the solution; m (mg) is the weight of the adsorbent used in every test; and k_i and k_j are the distribution coefficients of Ag(I) and the co-existing metal ions, respectively.

3. Results and discussion

3.1 Degree of grafting

As shown in Fig. 1(a), by varying the concentration ratio of GMA to SSS from 0 to 1.91 in binary monomer mixed solutions, the total DG increases and reaches a plateau at a concentration ratio of 1.32 to 1.47, followed by a boost, which is a common phenomenon in mutual radiation grafting polymerization.²⁵ With an increase in the concentration ratio of GMA and SSS, the process of the change in the grafting

degree can be divided into three stages, and a schematic diagram of the speculated grafting polymerization at different concentration ratio ranges of GMA and SSS is shown in Fig. 2.

When the concentration of GMA is lower than that of SSS (step 1 in Fig. 1(a)), it can be seen that SSS is mainly grafted from a binary mixture of monomers onto a PP matrix (Fig. 2(a)). Upon a further increase in the GMA concentration, the grafting of SSS increases slowly, while the increasing trend of total DG consistent with that of GMA (step 2 in Fig. 1(a)). This may be because the introduction of GMA in solvent can improve the surface polarity of non-polar PP nonwoven and promote the affinity between the SSS and nonwoven. When the GMA concentration is low (ratio < 0.60), SSS is preferentially grafted to the PP matrix. As SSS is grafted onto the PP substrate, the polarity of the PP substrate increases gradually, which further enhances the affinity between the solution containing SSS and PP resulting in an increase in the SSS grafting. With an increase in the GMA content, the DG of GMA increases faster than that of SSS and finally, the DG of the two monomers are close to each other (Fig. 2(b)). In the case where the GMA concentration is higher than that of SSS (ratio > 1.0), the GMA monomer will compete with SSS for free radical active sites on the nonwoven, therefore the DG of SSS tends to rise slowly. In a simultaneous radiation grafting process, if the reaction reaches a high DG (more than 100%), the grafting process is to a large extent initiated by the free radicals produced on the grafting branches (Fig. 2(c)), and this is the reason why a boost in DG is observed in step 3 in Fig. 1(a).

On the other hand, it can be inferred from Fig. 1(b) that IEC decreases with an increase in the GMA (epoxy) content on the PP-g-SSS/GMA nonwovens, yet the result also indicates that the IEC and epoxy content can be straightforwardly regulated by adjusting the concentration ratio of the monomers in the initial solution. Furthermore, the IEC of the obtained PP-g-SSS/GMA nonwovens is generally higher than those of some commercial cation exchange nonwovens.^{1,22} Considering the coordination between the content of the cation exchange groups (–SO₃Na) and the modified epoxy

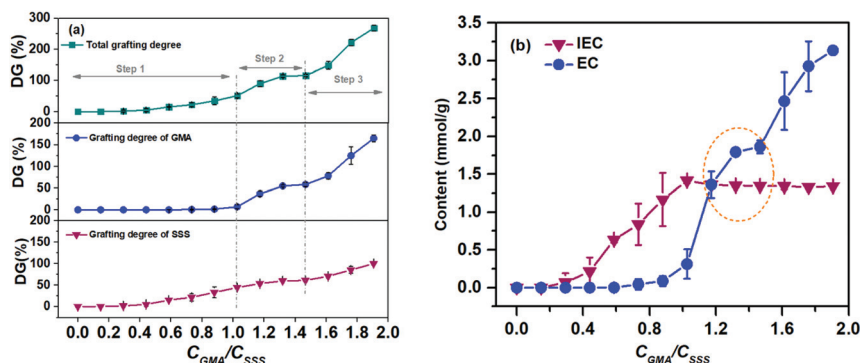


Fig. 1 (a) Effects of the concentration ratios of GMA and SSS in the simultaneous irradiation-grafting system on the total degree of grafting and degree of grafting of SSS and GMA at an adsorbed dose of 5 kGy. (b) IEC and EC values of PP-g-SSS/GMA corresponding to different concentration ratios.

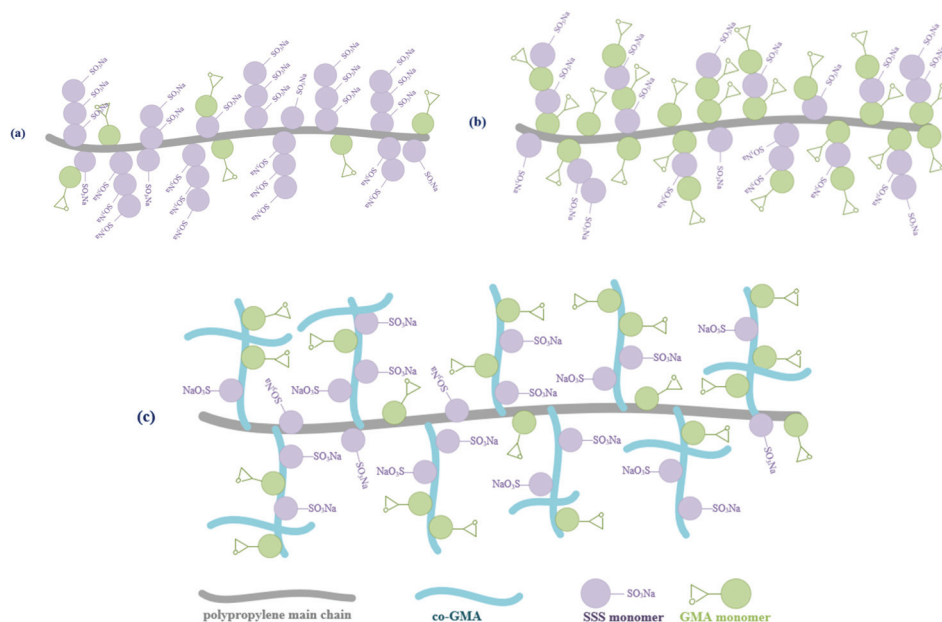


Fig. 2 Schematic diagrams of the speculated graft polymerization corresponding to different concentration ratios of GMA and SSS: (a) 0–1.03; (b) 1.03–1.46; and (c) 1.46–1.91.

groups on the polymer, the optimal concentration ratio of GMA and SSS in the reaction system is in the range of 1.17–1.46 in this study.

3.2 Characterization and properties of graft copolymerization

The structural investigations of pristine PP nonwovens before and after radiation grafting, as well as confirmation of the hydrophilic performance of the copolymer were unambiguously accomplished using scanning electron microscopy (SEM), energy-dispersive X-ray spectroscopy (EDS), attenuated total reflectance (ATR) spectroscopy, thermogravimetric analysis (TGA), differential scanning calorimetry (DSC), and contact angle measurements. The results clearly demonstrate that GMA and SSS were successfully grafted onto PP nonwovens and that the hydrophobicity of the nonwoven fabrics was successfully reversed.

3.2.1 Surface morphology and EDS. In this work, PP-*g*-SSS/GMA materials with different DG were obtained *via* simultaneous γ -ray induced radiation grafting polymerization with different initial GMA concentrations and their morphologies were found to strongly change compared with that of pristine PP nonwoven. As shown in Fig. 3(a) and (b), the fibers of PP nonwoven are smooth with an intertwined structure. The average diameter of the pristine PP fibers is approximately 25 μm . After modification, the fibers were thickened from 28 to 35 μm , with an increase in the DG from 26% to 220% and the surface of the fibers shows a tendency to be rougher as well.

EDS analysis was applied to study the elemental composition on the nonwovens surface corresponding to different DG. One typical point was selected on each nonwoven and energy spectra were obtained. Compared with pristine PP non-

woven (k), the energy spectra ((l), (m), (n), (o)) of all of the PP-*g*-SSS/GMA samples show evident peaks for sulfur, sodium, and oxygen. This result suggests that sulfonic acid groups were successfully introduced to the PP nonwovens. Besides this, the higher oxygen peaks could also indicate the introduction of epoxy and carboxylic acid groups. The chemical structures of the modified nonwovens were further investigated *via* ATR spectra analysis.

3.2.2 Chemical structure analysis. In order to investigate the graft copolymer formation, ATR studies were carried out. The infrared spectra of PP and the modified PP-*g*-SSS/GMA compositions with different DG are shown in Fig. 4. For the spectra of the original PP nonwoven, the adsorption bands at 2915 and 2840 cm^{-1} can be attributed to the asymmetric and symmetrical stretching vibrations of $-\text{CH}_2$, respectively. The peak at 1448 cm^{-1} can be attributed to an asymmetrical $-\text{CH}_3$ bending vibration, while the peak at 1375 cm^{-1} corresponds to a symmetrical $-\text{CH}_3$ bending vibration.²²

For PP-*g*-@GMA/SSS, these aforementioned bands are present in its spectrum. In addition, the first new broad adsorption band between 3100 to 3660 cm^{-1} represents the stretching vibration of H_2O , which confirms that the modified nonwovens are hydrophilic. The peaks at 754 cm^{-1} and 681 cm^{-1} are due to an out-of-plane $-\text{C}-\text{H}$ (in aromatics) bending vibration, which confirms the presence of a benzene ring that belongs to SSS. In addition, peaks owing to $-\text{SO}_3$ are observed at 1124 and 1173 cm^{-1} for asymmetric stretching vibrations and 1035 and 1010 cm^{-1} for symmetric stretching vibrations.^{23,26,27} Moreover, the spectra of the samples show two peaks in the absorption band from 1200 to 1070 cm^{-1} , indicating the presence of the $\text{C}-\text{O}$ stretching vibration of the ester group in addition to the $\text{S}-\text{O}$ stretching vibration of the

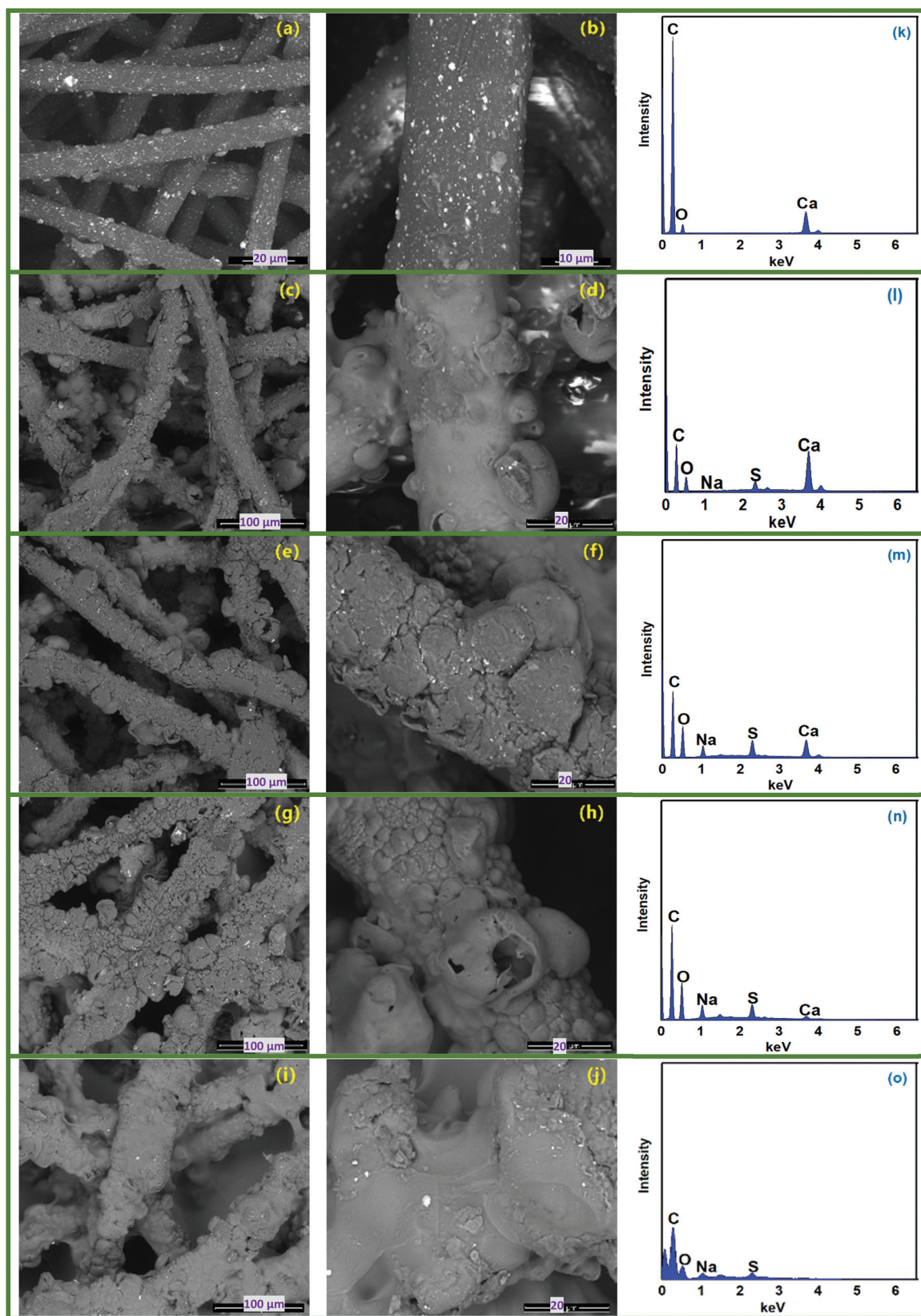


Fig. 3 (a and b) SEM images of pristine PP nonwoven and PP-g-GMA/SSS with different DG: (c and d) PP-g-GMA/SSS-56, (e and f) PP-g-GMA/SSS-86, (g and h) PP-g-GMA/SSS-118, and (i and j) PP-g-GMA/SSS-220. EDS elemental composition analysis of (k) pristine PP nonwoven, (l) PP-g-GMA/SSS-56, (m) PP-g-GMA/SSS-86, (n) PP-g-GMA/SSS-118, and (o) PP-g-GMA/SSS-220. The number following PP-g-GMA/SSS indicates the DG.

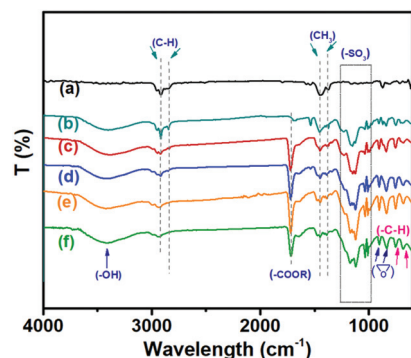


Fig. 4 ATR spectra of (a) pristine PP nonwoven, (b) PP-g-SSS/GMA-26, (c) PP-g-SSS/GMA-56, (d) PP-g-SSS/GMA-86, (e) PP-g-SSS/GMA-118, and (f) PP-g-SSS/GMA-220. The number following PP-g-SSS/GMA indicates the DG.

sulfonic acid group. The next remarkable peak at 1719 cm^{-1} is due to the -C=O stretching vibration of the ester group. Besides this, new characteristic peaks belonging to the stretching vibration of the epoxy group at 905 cm^{-1} and 841 cm^{-1} ascertain the successful grafting of GMA.²⁷ The DG of the different samples is different, but there is little difference in the areas of the infrared absorption peaks of the epoxy and sulfonic acid groups in spectra (c), (d), (e), and (f). This proves that radiation graft polymerization takes place not just on the surface, but inside the fibers.²⁸

3.2.3 Thermal properties

TGA. TGA curves of the parent PP and grafted PP-g-SSS/GMA nonwovens under N_2 are shown in Fig. 5. PP demonstrates excellent thermo-stable properties, decomposing at high temperature ($\sim 360\text{ }^\circ\text{C}$), which can also be observed in the DTG curve. And the weight loss is confined to a single-step degradation pattern.²⁹ There are three stages of weight loss in the case of PP-g-SSS/GMA. The first stage at $100\text{ }^\circ\text{C}$ shows a small peak before the melting peak, which is the dehydration stage where absorbed water is lost.¹³ This indicates that the irradiated samples exhibit strong water absorption, which is related to the successful introduction of SSS.^{28,30} Irradiated samples are thermally stable up to about $230\text{ }^\circ\text{C}$ and then start

degrading slowly up to $360\text{ }^\circ\text{C}$, which is attributed to the thermal decomposition of sulfonic and epoxy groups.^{31,32} Then, there is a major weight loss ascribed to the bulk decomposition of the polymer residue up to $446\text{ }^\circ\text{C}$, corresponding to the last peak in the DTG. The fractional mass loss before $450\text{ }^\circ\text{C}$ increased with increasing DG, a clear indication that the thermal decomposition and observed weight losses at lower temperatures were due to the GMA and SSS grafting.³³

PP and grafting samples exhibit earlier weight losses with the thermal treatment temperature under air conditions (Fig. 6) as compared with the samples treated under a N_2 atmosphere. In an oxidative environment, the degradation of samples starts at about $200\text{ }^\circ\text{C}$. The TGA curves indicate that the radiated-grafted non-woven samples still maintain great thermal stability at $200\text{ }^\circ\text{C}$ under both nitrogen and air conditions and the DG does not significantly alter the decomposition temperature of the samples.

DSC. The pristine and irradiated PP and PP-g-SSS/GMA samples were also characterized to assess their thermal properties *via* DSC analysis. The DSC curve of pristine PP nonwoven shows endothermic melting at a melting temperature at $165.21\text{ }^\circ\text{C}$ due to the crystal structure present in this polymer (Fig. 7). The DSC curves for the PP-g-SSS/GMA-26 and PP-g-SSS/GMA-118 also present the same profile, with melting tempera-

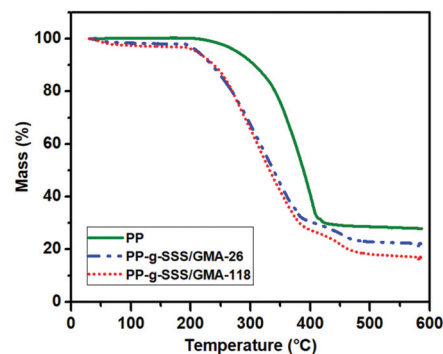


Fig. 6 TGA of PP nonwovens and irradiated PP-g-SSS/GMA with 26% DG and 118% DG under an air atmosphere, where the number following PP-g-SSS/GMA indicates the DG.

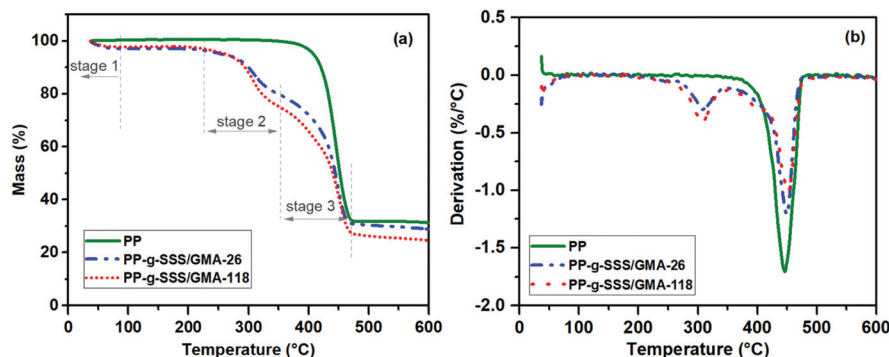


Fig. 5 (a) TGA and (b) DTG diagrams of the PP nonwovens and irradiated PP-g-SSS/GMA with 26% DG and 118% DG under a nitrogen atmosphere, where the number following PP-g-SSS/GMA indicates the DG.

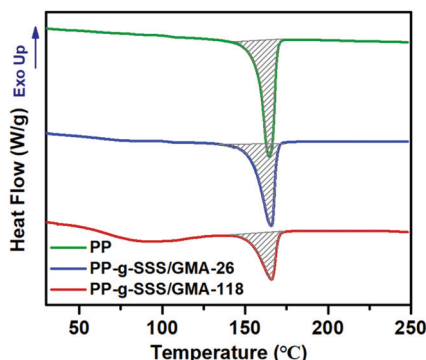


Fig. 7 DSC analyses of pristine PP non-woven fabric and irradiated PP-g-SSS/GMA with 26% DG and 118% DG, where the number following PP-g-SSS/GMA indicates the DG.

tures of 165.35 and 165.55 °C respectively, which indicates that the perfection of the crystallites is not influenced by the irradiation process under our experimental conditions.

The melting enthalpy (ΔH) of pristine PP was found to be 56.58 J g⁻¹, while the ΔH values of PP-g-SSS/GMA-26 and PP-g-SSS/GMA-118 decreased to 47.38 J g⁻¹ and 26.71 J g⁻¹, respectively. According to eqn (6), the DC of pristine PP is 27.33%. PP-g-SSS/GMA samples with 26% and 118% DG exhibit 22.89% and 12.90% DC. Indeed, after grafting polymerization with GMA and SSS, the decrease in DC of PP-g-SSS/GMA is caused by the introduction of amorphous graft chains which dilute the crystalline zone fraction of the PP matrix.³⁴ Eqn (7) was used to calculate the melting enthalpy of the PP components in the grafted samples. According to eqn (7), the $\Delta H_{f(PP)}$ values of PP-g-SSS/GMA-26 and PP-g-SSS/GMA-118 are 59.70 and 58.22 J g⁻¹, respectively, which are close to the 56.58 J g⁻¹ melting enthalpy of the PP matrix. These phenomena indicate that the SSS/GMA grafts are amorphous in nature and do not disrupt the inherent crystalline structure of the PP nonwoven matrix during the grafting process.³⁵

3.2.4 Contact angles. Fig. 8 shows the effect of different DG on the water contact angles of the pristine and grafted PP nonwovens. The PP matrix, being a hydrophobic polymer, pre-

sents a high contact angle with water (100.96°). As SSS is a hydrophilic monomer, the grafted polymers are relatively more hydrophilic than the pristine sample. From Fig. 8, it is obvious that the shapes of the water droplets created on the nonwovens experience fluctuating changes with different DG. The hydrophilicity of PP nonwovens before and after modification is revealed by the changes in the water contact angles. The contact angle values decrease from 100.96° to 96.66°, and 0° with increasing DG corresponding to (a), (b), (c), (d), and (e), respectively. In combination with Fig. 1(a), the growth of hydrophilicity is related to the increase in the increasing introduction of SSS on the PP nonwovens.

However, the hydrophilicity of PP-g-SSS/GMA worsens as the DG increases from 118% to 220%, where the DG of GMA increases dramatically. This may be due to the grafting of a large amount of GMA on the nonwoven surface, making it inaccessible for the water droplets to make contact with the hydrophilic sulfonic groups. This result is in good agreement with the results revealed by the other characterization methods.

The results indicate that the obtained PP-g-SSS/GMA nonwovens with DG of 56% to 118% show superhydrophilic properties, corresponding to concentration ratios of monomer GMA and SSS within the range of 1.03–1.46 as discussed above; at the same time, this result is consistent with the concentration ratio range corresponding to the optimal cation exchange capacity and epoxy content as discussed above. On the other hand, the radiation-induced simultaneous grafting product PP-g-GMA/SSS has potential applications for use in water treatment.

3.2.5 Mechanical properties. Mechanical properties are of great importance for PP fibrous materials from a practical standpoint. Tensile properties are expected to be maintained to a large extent after grafting modification. The tensile strength (σ_b , MPa) values of the original PP nonwovens and grafted samples PP-g-SSS/GMA with different DG are shown in Table 1. The tensile strength was about 3.0 MPa for pristine PP and decreased with an increase in DG. Before the DG of 86%, the tensile strength of the samples was maintained at above 2.0 MPa, where the decrease in the tensile strength of the

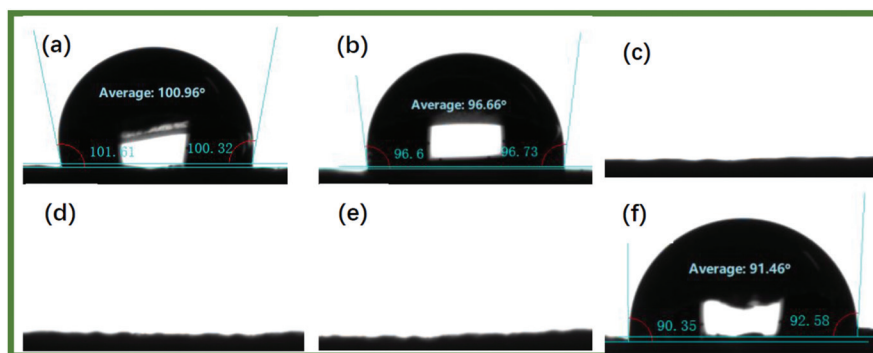


Fig. 8 Measurements of the water contact angles of (a) pristine PP, (b) PP-g-SSS/GMA-26, (c) PP-g-SSS/GMA-56, (d) PP-g-SSS/GMA-86, (e) PP-g-SSS/GMA-118, and (f) PP-g-SSS/GMA-220, where the number following PP-g-SSS/GMA indicates the DG.

Table 1 Tensile strengths of PP and PP-*g*-SSS/GMA with different DG values

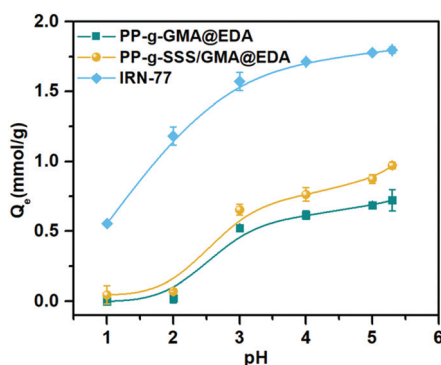
Sample	σ_b (MPa)
PP	3.00 ± 0.11
PP- <i>g</i> -SSS/GMA-26	2.19 ± 0.09
PP- <i>g</i> -SSS/GMA-56	2.18 ± 0.33
PP- <i>g</i> -SSS/GMA-86	2.04 ± 0.17
PP- <i>g</i> -SSS/GMA-118	1.67 ± 0.04
PP- <i>g</i> -SSS/GMA-220	1.58 ± 0.17

samples compared with the original PP was mainly caused by irradiation damage. When the grafting is high ($DG \geq 118\%$), the introduction of too many grafts has a negative effect on the mechanical properties of the material. Therefore, combining ICE, EC and other parameters, PP-*g*-SSS/GMA-86 was selected as the experimental objective for further modification in the following application tests.

3.3 Ag(I) Adsorption tests

After successful modification with EDA, both PP-*g*-SSS/GMA and PP-*g*-SSS/GMA reached 98% conversion, resulting in 1.18 mmol EDA per g polymers. The adsorption of Ag(I) was used to determine the application performance of the adsorbents, in which the selectivity and adsorption rate are important evaluation parameters. In a comparative study of the three adsorbents PP-*g*-GMA@EDA, PP-*g*-SSS/GMA@EDA and IRN77 in Ag(I) adsorption, it was found that PP-*g*-SSS/GMA@EDA and PP-*g*-SSS/GMA@EDA show great selection towards Ag(I) due to the introduction of amine groups, and faster adsorption benefiting from the presence of sulfonic groups. This also further confirms the applicability of the bifunctional material PP-*g*-SSS/GMA in this study in water treatment.

3.3.1 Effect of solution acidity on Ag(I) adsorption and selectivity. Fig. 9 represents the effect of the initial pH of the solution on Ag(I) adsorption. According to Fig. 9, the Ag(I) adsorption capacity of the three adsorbents increases along with an increase in the pH value, among which IRN77 shows the highest Ag(I) adsorption capacity. This is because the competition of H^+ with Ag(I) in the liquid phase for adsorption

**Fig. 9** Effect of the initial solution acidity (pH) on the adsorption of Ag(I) by PP-*g*-GMA@EDA, PP-*g*-SSS/GMA@EDA, and IRN77.

sites reduced as the solution acidity decreased (as the pH value increased). In addition, the deprotonation of the amine groups on the adsorbent PP-*g*-GMA@EDA and PP-*g*-SSS/GMA@EDA improved the adsorption capacity of Ag(I).

Fig. 10 illustrates the adsorption capacity of PP-*g*-GMA@EDA and IRN77 towards different metal ions in the mixed solution of Ag(I) and coexisting ions Mn(II), Co(II), Ni(II) and Sr(II) at different acidity, and the separation factors of Ag(I) with every coexisting metal ion were calculated and are shown in Table 2. Although IRN77 has an advantage in Ag(I) adsorption in a single metal ion solution, it exhibits worse Ag(I) selectivity in a mixed solution of metal ions, while it exhibits strong adsorption of Sr(II). This means that Ag(I) competes less effectively with co-existed divalent metal ions for the reason that the affinity of the sulfonic acid adsorbent for various common cations increases with increasing cation charge.³⁶ In contrast, the adsorbent PP-*g*-SSS/GMA@EDA containing amine groups shows very high adsorption selectivity towards Ag at both pH3 and pH5.3, as amine groups have a better affinity towards Ag(I) rather than the other metal ions.

3.3.2 Adsorption kinetics. Fig. 11 shows a comparison of the Ag(I) adsorption capacity of PP-*g*-GMA@EDA and PP-*g*-SSS/GMA@EDA as a function of contact time, which is an essential parameter for economical wastewater treatment process. The difference between these two adsorptions lies in the presence of the sulfonic acid groups. The data clearly shows that in the first 100 minutes, the Ag(I) adsorption of PP-*g*-SSS/GMA@EDA increased rapidly, and the adsorption rate was almost twice that of PP-*g*-GMA@EDA. After that, the adsorption of both tends to slowly grow at similar rates. Equilibrium was reached at 360 and 420 minutes for PP-*g*-SSS/GMA@EDA and PP-*g*-GMA@EDA, respectively, where the maximum adsorptions of 0.81 and 0.59 mmol g⁻¹ take place. PP-*g*-SSS/GMA@EDA not only shows a rapid initial adsorption rate, but also reaches adsorption equilibrium 1 h earlier than PP-*g*-GMA@EDA without any sulfonic acid groups.

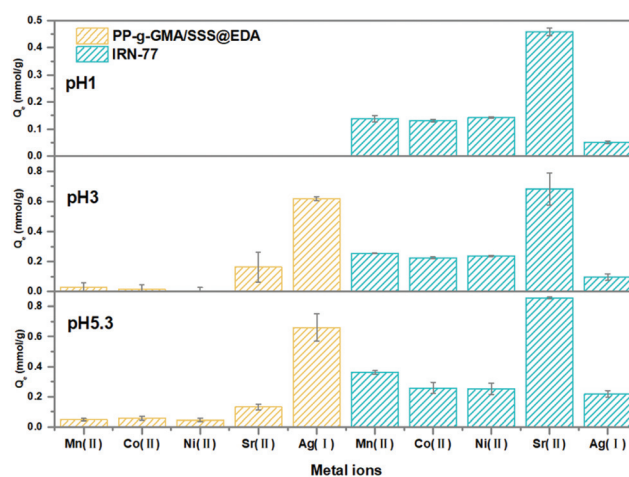
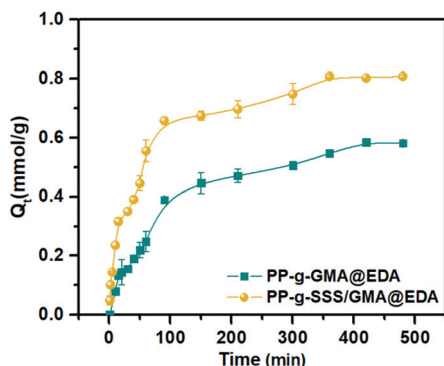
**Fig. 10** Adsorption capacities of PP-*g*-GMA@EDA and IRN77 towards different metal ions in solution mixtures of Ag(I) and the coexisting metal ions Mn(II), Co(II), Ni(II), and Sr(II) at pH 1, 3, and 5.3.

Table 2 Separation factors of Ag(I) toward each co-existing metal ion in solution mixtures with different pH values adsorbed by PP-g-SSS/GMA@EDA and IRN77

Ag(I)-coexisting ions	pH 1.0 SF (IRN77)	pH 3.0		pH 5.3	
		SF (PP-g-SSS/GMA@EDA)	SF (IRN77)	SF (PP-g-SSS/GMA@EDA)	SF (IRN77)
Ag(I)-Mn(II)	0.34	58.84	0.31	39.96	0.49
Ag(I)-Co(II)	0.36	93.76	0.37	32.78	0.80
Ag(I)-Ni(II)	0.32	905.23	0.34	41.66	0.82
Ag(I)-Sr(II)	0.06	9.66	0.05	14.48	0.05

**Fig. 11** Effects of contact time on the adsorption of Ag(I) by PP-g-GMA@EDA and PP-g-SSS/GMA@EDA.

To better explore the adsorption kinetics of the adsorbents towards Ag(I) and the diffusion mechanism of the adsorbents, pseudo-first-order (eqn (12)), pseudo-second (eqn (13)) and intraparticle diffusion (eqn (14)) models were used to analyze the experimental data obtained from contact time studies.^{20,37}

$$\ln(Q_e - Q_t) = \ln Q_e - k_1 t \quad (12)$$

$$\frac{t}{Q_t} = \frac{1}{k_2 Q_e^2} + \frac{t}{Q_e} \quad (13)$$

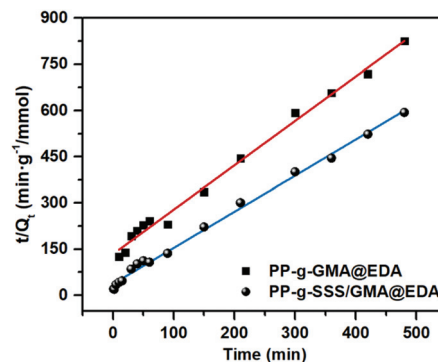
$$Q_t = k_{id} t^{1/2} + C \quad (14)$$

where Q_t and Q_e are the amounts of Ag(I) adsorbed (mmol g^{-1}) at any instant time t (min) and at equilibrium, respectively, k_1 and k_2 are the pseudo-first-order pseudo-second-order rate constants, respectively, k_{id} is the intraparticle diffusion rate constant ($\text{mg g}^{-1} \text{min}^{-1/2}$) and C is the thickness of the boundary layer.

The modeled kinetic parameters and regression correlation coefficients (R^2) were calculated and the data are shown in Table 3. As shown in Table 3, a higher R^2 value (0.99) clearly

indicates the validity of the pseudo-second-order model to fit the experimental data corresponding to the adsorbents PP-g-SSS/GMA@EDA and PP-g-GMA@EDA (Fig. 11). And the adsorption capacity value obtained by pseudo second-order modeling is in good agreement with the experimental data (Fig. 12).

The parameters related to the intraparticle diffusion model are summarized in Table 4. The higher value of k_{id} for PP-g-SSS/GMA@EDA than that for PP-g-GMA@EDA indicates its higher initial rate of Ag(I) adsorption. The intraparticle diffusion plots of both adsorbents show multi-linearity in the process, indicating that three steps exist, as shown in Fig. 13. The first step is the diffusion of adsorbate through the solution on the external surface of the adsorbent or the boundary

**Fig. 12** Pseudo-second-order fitting curves of Ag(I) adsorption onto PP-g-GMA@EDA and PP-g-SSS/GMA@EDA.**Table 4** Intraparticle diffusion parameters for the adsorption of Ag(I) onto PP-g-GMA@EDA and PP-g-SSS/GMA@EDA

Adsorbent	k_{id} ($\text{mmol g}^{-1} \text{min}^{-1/2}$)	C (mmol g^{-1})	R^2
PP-g-GMA@EDA	0.030	0.012	0.95
PP-g-SSS/GMA@EDA	0.035	0.15	0.88

Table 3 Kinetics parameters for the adsorption of Ag(I) onto PP-g-GMA@EDA and PP-g-SSS/GMA@EDA

Adsorbent	Pseudo-first-order kinetics model			Pseudo-second-order kinetics model		
	k_1 (min^{-1})	Q_e (mmol g^{-1})	R^2	k_2 ($\text{g} \cdot \text{mmol}^{-1} \text{min}^{-1}$)	Q_e (mmol g^{-1})	R^2
PP-g-GMA@EDA	0.0090	0.58	0.93	0.015	0.69	0.99
PP-g-SSS/GMA@EDA	0.0083	0.58	0.90	0.037	0.85	0.99

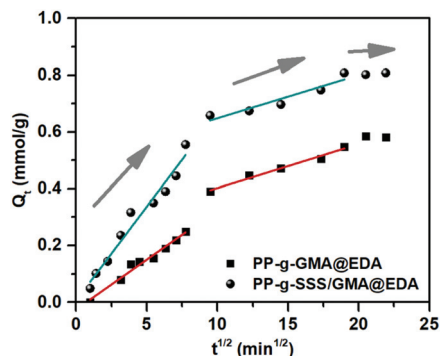


Fig. 13 Intraparticle diffusion fitting curves for Ag(I) adsorption onto PP-g-GMA@EDA and PP-g-SSS/GMA@EDA.

surface diffusion of the sorbate molecules. The second step shows gradual adsorption, where intraparticle diffusion is the rate-limiting factor, and the third step is related to the final equilibrium achieved. The three-step process shows that the adsorption process occurs *via* surface adsorption and intraparticle diffusion.³⁸ It is worth noting that in the first step, PP-g-SSS/GMA@EDA exhibits a through-origin curve with a steeper slope (a better k_{id} value), which means that the mass transfer rate is the determinant factor of the rate limiting, and that the adsorption rate is relatively high. This could be due to the presence of sulfonate groups on the adsorbent as they promote mass transfer between the solid and liquid phases.

4. Conclusions

This research provides a straightforward way to synthesize highly hydrophilic PP-g-SSS/GMA nonwovens containing epoxy and cation exchange groups (sulfonic groups). GMA and SSS as functional monomers are grafted onto PP nonwovens *via* simultaneous radiation-induced graft polymerization. This methodology is also an alternative green way of synthesizing cation exchange nonwovens with further modifiable epoxy groups.

The effects of the content ratio of monomers in a radiation graft system on DG have been studied. Combining the structural and chemical characterization of the PP-g-SSS/GMA composites, it was found that GMA and SSS were successfully grafted not only on the surface of PP nonwovens but also in the interior amorphous region of the fibers. By carrying out thermodynamic characterization, the new composites proved to be thermally stable at over 200 °C.

It was established that the radiation-induced graft copolymer PP-g-SSS/GMA is a bifunctional composite. It was found that the IEC and epoxy content could be adjusted *via* changing the concentration ratio of GMA and SSS in the initial mixed solution. The highest ion exchange capacity of the nonwovens was above 1.5 mmol g⁻¹. Besides this, *via* further modifying the epoxy group, the produced graft copolymer can be used for preparing adsorbents that exhibit strongly hydrophilic properties and high adsorption rates for the selective removal of

metal ions in aqueous solution, which was experimentally confirmed based on Ag(I) adsorption. Thus, the obtained PP-g-SSS/GMA material shows potential applicability as an active part of devices used in water treatment.

Author contributions

Xiao-han Pan: conceptualization, methodology, validation, writing – original draft and visualization. *Jian-hua Zu: writing – review and editing, supervision, project administration and funding acquisition.

Conflicts of interest

The authors declare no conflicts of interest.

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